

Statistical Inference and Modeling

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- Lecture 3: Resampling
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- Lecture 5: Robust regression
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1. Introduction

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Prerequisite

This class focuses on

- selected techniques used in supervised and unsupervised learning
- analysis of statistical properties of models

For these reasons, you should have some basic skills on

- Statistics to analyze all random quantities;
- Mathematics on linear algebra, differential equations, and calculus;
- Optimization to understanding the learning of model in training process;
- Python coding: <https://github.com/GenAI-CUEE/Statistical-Learning-EE575-Y2024>.

Objective of this course

- Know how to extract the properties and use the right measurements to visualize the data.
- Can choose the suitable statistical tools/machine learning for solving a given problem.
- Solve given tasks and some real-world problems with statistical and machine learning.
- Analyze problems in the statistical tools and machine learning algorithms.

Activities and scores:

- Homeworks 50% (5 homeworks).
- Exam (50%) - 25% midterm exam and 25% final exam.

What is 'making an inference'?

Suppose that we have a system that gives us an image of a desired pet. Let's find out the output of system $y = f(x)$ by 'making an inference'?

discriminative tasks → size ของ output จำกัด (solution จำกัด)

x_j



$$y_j = f(x_j)$$

hamster

pony

kitty

parrot

What is 'making an inference'?

Suppose that we have a system that gives us an image of a desired pet. Let's find out the output of system $y = f(x)$ by 'making an inference'?

generative tasks $\rightarrow$ continuous space กว้าง.

x_j

$x_1 = \text{a pet animal}$

$x_2 += \text{whiskers \& fur}$

$x_3 += \text{vegetarians}$

$x_4 += \text{long ears}$

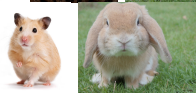
$y_1 \in$



$y_2 \in$



$y_3 \in$



$y_4 =$



เพิ่มรายละเอียดขึ้นเรื่อยๆ $\Rightarrow$ Set of solution ระบุแคลง

What is Statistical Learning

Data components in learning

ตัวแปรต้น, ตัวแปรอิสระ

เราจะใช้คำนี้ตลอด.

- The input: X is the **independent** variables (predictors, input features).
- The output: $\hat{Y}$ is the **dependent** variables (response, prediction, predicting label, predicting target)
ตัวแปรตาม
- The reference:
 - y_o is the **noiseless** reference (the ground truth label or the ground truth target). $\rightarrow$ Ideal case
 - Y is the **noisy** reference (collected data/ measurements from sensors).

The relationship between X and $\hat{Y}$ is represented in the following form:

$$\hat{Y} = f(X) \rightarrow f(X; \theta)$$

feature parameter
hyper
เราไม่ใช้เรียลสตอนเขียน
แต่ให้ keep in mind ว่ามีด้วย

where $f(\cdot)$ is an **unknown function** that extracts the information that X provides about $\hat{Y}$.

↑
เป้าหมายหลัก.

Statistical learning refers to approaches for estimating Y with $\hat{Y}$ [2]:

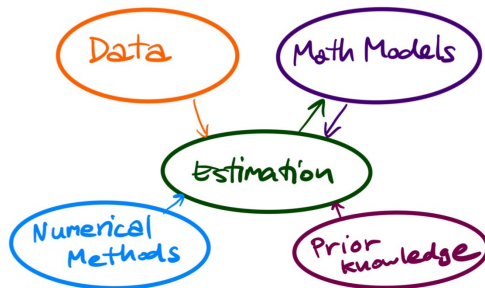
$$\hat{Y} \approx Y = y_o + \mathcal{E}$$

↑
ในทางทฤษฎี เราจะพยายามมองว่า
 $\mathcal{E} \approx 0$ สำหรับการ modeling.

where $\mathcal{E}$ is some stochastic **error** in the system.

What is Statistical Learning

users develop a math model to explain data using prior knowledge of applications



- applicable estimation techniques depend on a selected model
- most model estimation problems require numerical methods to get a numerical solution

How to derive $f(x)$ from data ?

Therefore, there are 3 steps.

- Training step: ★★

This step is used in finding θ and $f(\cdot; \theta)$.

- Validation step: ★ / Validation data.

This step aims at finding the hyperparameters of the trained model.

- Testing step: / Inference phase.

this happens during applying $f(\cdot; \theta)$ for deployment

6 Types of models

1. Parametric and non-parametric methods

Definition

Common assumptions

Advantages/disadvantages

Examples

2. Learning Methods

Supervised learning

Unsupervised Learning

Self-supervised learning

3. Classifying Models in Bayesian Context $\Rightarrow$ Generative vs. Discriminative

Parametric and non-parametric methods: definition

Parametric methods

- Statistical techniques that rely on **specific assumptions** about the underlying distribution of the population being studied.
- There is a set of **fixed parameters** that are used to determine a probability model that is used in Machine Learning as well.
- Those methods for which we priory know that the population is **normal distribution**.

Non-parametric methods

- Statistical techniques that **do not rely on specific assumptions** about the underlying distribution of the population being studied.
- **No need to make any assumption** of parameters for the given population or the population we are studying.
- They are **distribution-free** because they make no assumptions about the shape of the distribution.

The content is adopted from [1].

Parametric and non-parametric methods: common assumptions [1]

Parametric methods

- Normality: The data follows a normal (Gaussian) distribution.
- Homogeneity of variance: The variance of the population is the same across all groups.
- Independence: Observations are independent of each other.

Non-parametric methods

- Independence: Data points are independent and not influenced by others.
- Random Sampling: Data represents a random sample from the population.
- Homogeneity of Measurement: Measurements are consistent across all data points.

Parametric and non-parametric methods: advantages/disadvantages [1]

Parametric methods

- (+) More powerful when the assumptions are met.
- (+) More efficient.
- (+) Provide the estimates of population parameters.
- (−) Sensitive to assumption.
- (−) Limited flexibility.
- (−) May not capture complex relationship.

Non-parametric methods

- (+) Robust to outliers.
- (+) Widely applicable.
- (+) Easy to implement.
- (−) Less powerful when the assumption of the parametric methods are met.
- (−) May require larger sample sizes.
- (−) Less information about the population.

Parametric and non-parametric methods: examples [1]

Parametric methods $\rightarrow$ MLE, MoM, ... $\Rightarrow$ Assume formulation : X vs. θ

- **Linear regression:** Predicts a continuous outcome based on a linear relationship with one or more independent variables.
- **Logistic regression:** Predicts a binary outcome (e.g., yes/no) based on a set of independent variables.
- **Naive Bayes:** Classifies data points based on Bayes' theorem and assuming independence between features. including LDA, QDA
- **Hidden Markov Models:** Models sequential data with hidden states and observable outputs.

Non-parametric methods $\Rightarrow$ data-driven $\hat{f}(\cdot)$ without Assume.

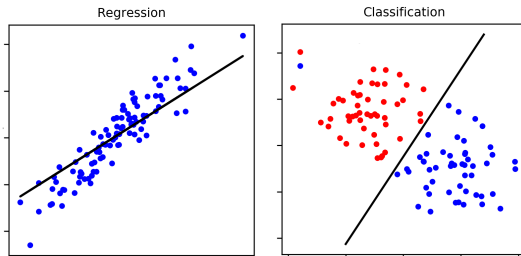
- **K-Nearest Neighbors (KNN):** Classifies data points based on the k nearest neighbors.
- **Decision Trees:** Makes classifications based on a series of yes/no questions about the features.
- **Support Vector Machines (SVM):** Creates a decision boundary that maximizes the margin between different classes.
- **Neural networks:** Can be designed with specific architectures to handle non-parametric data, such as convolutional neural networks for image data and recurrent neural networks for sequential data.

Supervised Learning

- Training data: a set containing pairs of input information and labels $(x, y) \in (\mathcal{X}, \mathcal{Y})$
- The learning is aimed at finding the most suitable model parameter $\hat{\theta}$ for the following problem.

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} (f(x; \theta) - \overset{\hat{y}}{y})^2$$

Examples: Regression, Classification, Support Vector Machine.



<https://www.engineersgarage.com/machine-learning-algorithms-classification/>

Unsupervised Learning

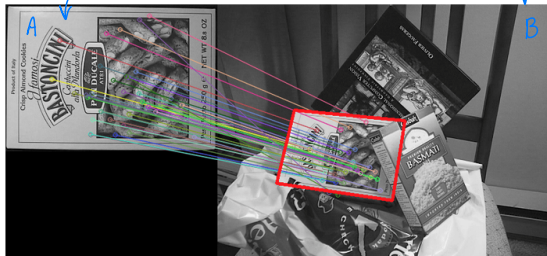
- Training data: a set of data without label.
- Unsupervised learning does not use data with explicit labeling. Thus, its learning is often designed for specific tasks.
- Local feature matching. Learn a model $f(x; \theta)$ that can map the same object between two images, e.g., x_+ and x_- from the same image set $\mathcal{X}$.
- In this case, $f(x; \theta)$ is often used to generate the representation vector of the object.
- Then, the learning is aimed at finding the parameters θ that enforces the following property:

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} f(x_+; \theta)^T f(x_-; \theta)$$

$$\hat{S}_+ = f(x_+; \theta)$$

$$\hat{S}_- = f(x_-; \theta)$$

$$S_+, S_- \in \mathbb{R}^n$$



$$n \times n$$

$$\underbrace{256 \times 256}_{[256]^2 \times 1}$$

$$\hat{S}_+^T \cdot \hat{S}_- \in \mathbb{R}^1$$

Learning Methods: self-supervised learning

Knowledge distillation : การสกัดข้อมูลจาก model ที่ถูกเทรนแล้ว.

Self-supervised learning

- Suppose that we have limited amount of data x with label y from $(\mathcal{X}^o, \mathcal{Y}^o)$.
- Train the prediction model $f(\cdot; \theta)$ using the input image x and y from $(\mathcal{X}^o, \mathcal{Y}^o)$.
- Then, the trained model is used to infer the label of $x \in \mathcal{X}^+$ to generate an unseen set of label $\mathcal{Y}^+$.



3. Classifying Models in Bayesian Context

หลายที่จะเขียน $\Rightarrow p(X|Y) \propto p(Y|X)p(X)$

Bayes' Theorem:

$$p(X|Y) = \frac{p(Y|X)p(X)}{p(Y)} \approx p(Y|X)p(X)$$

prior

Label (สิ่งที่เราจะ infer)

โดยทั่วไป เราจะ ignored มันไปก่อน



Y := Label, output
e.g. y = cat

Types of Models in Bayesian Context [3]

- **Generative Model.** $\Rightarrow p(X)$
- **Conditional Generative Model.** $\Rightarrow p(X|Y)$
- **Discriminative Model.** $\Rightarrow p(Y|X)$

The types of these models indicate which probability function is being used in learning the model parameters.

Generative Model. → วาดสิ่งที่ไม่เคยมี

Learn the distribution $p(X)$

↳ มักมีหลาย Solution.

→ สิ่งไหนวาดตามทีบอ

Conditional Generative Model.

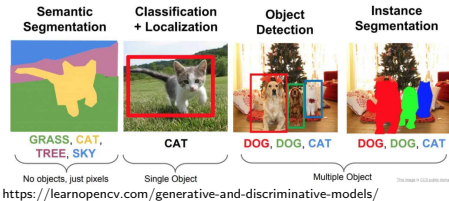
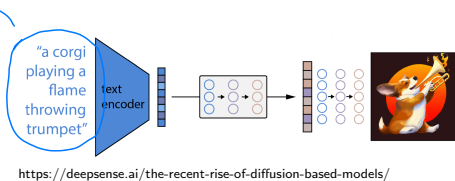
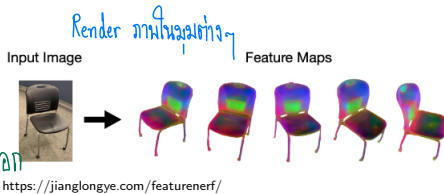
Learn the conditional distribution $p(X|Y)$.

→ ทำนายค่าตอบ

Discriminative Model.

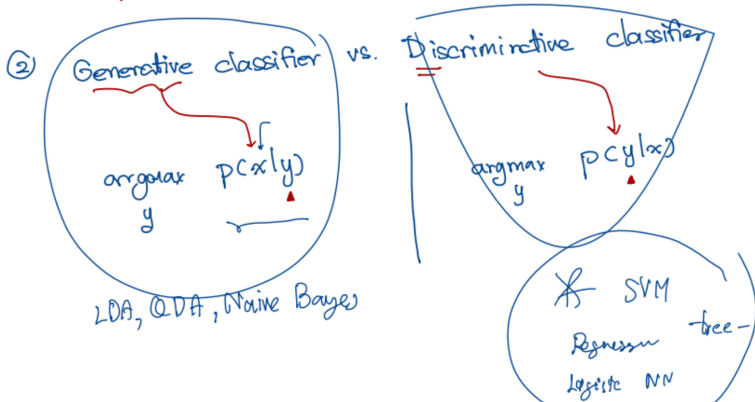
Learn the conditional distribution $p(Y|X)$

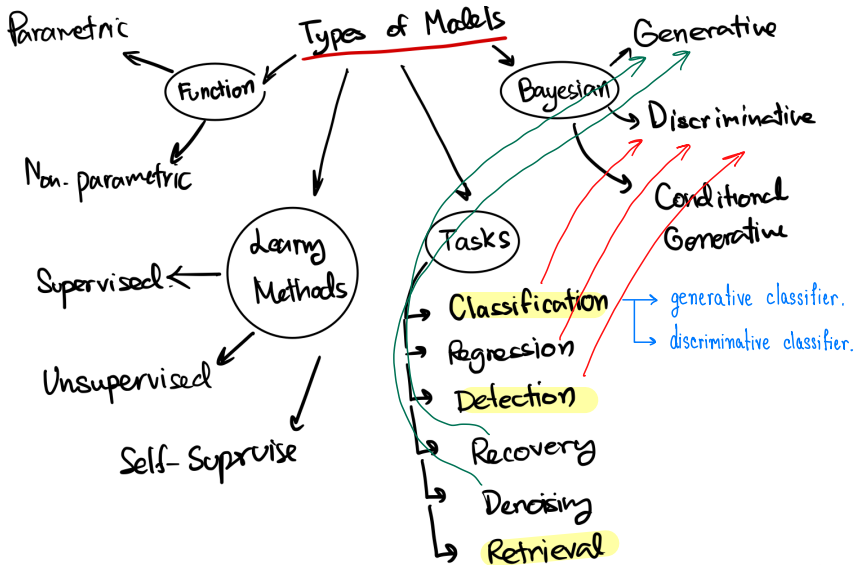
↳ มักมี 1 Solution * เราจะเรียนในวิชาขึ้น
เป้าหมายหลัก



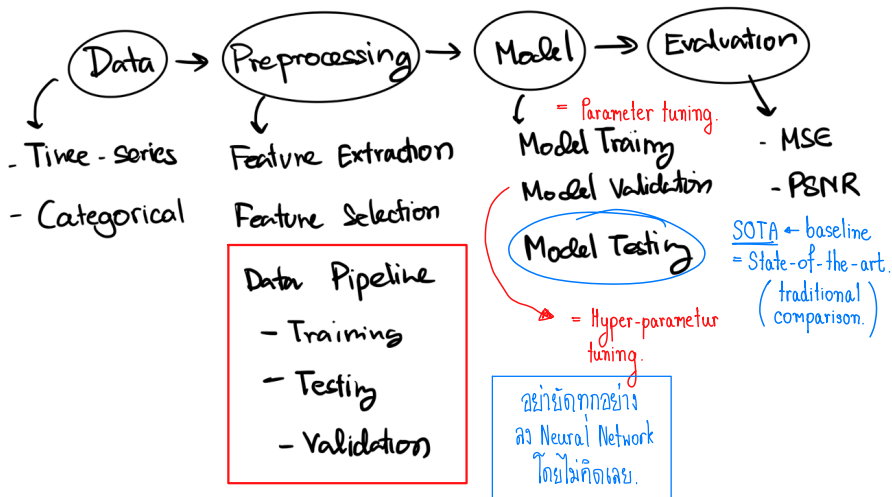
Generative Classifier vs. Discriminative Classifier

① All classifiers are a discriminative model.

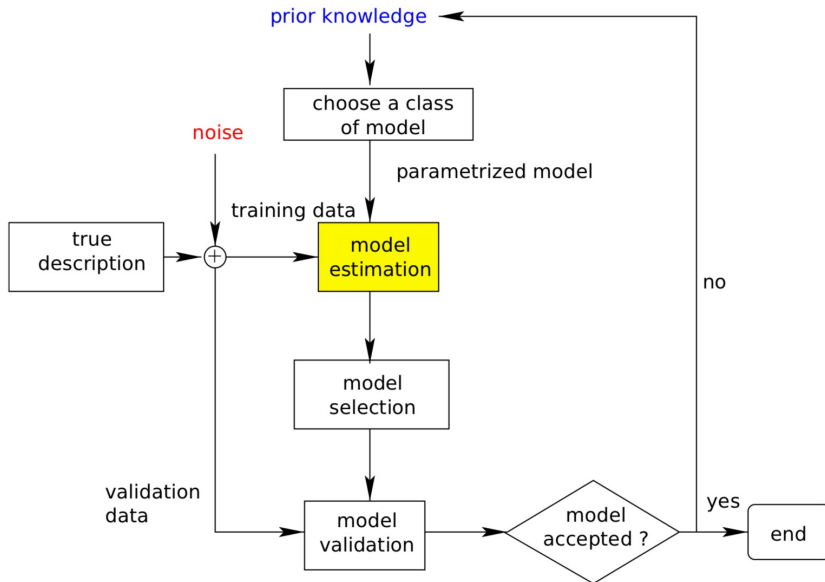




Pipeline



Pipeline



What we plan to learn

• Supervised learning →

- Linear regression
- Regression + regularization
- Neural network
- Logistic regression
- Naive Bayes
- K-nearest neighbors (k-NN)
- LDA, QDA
- Support vector machine
- Tree-based methods

Object tracking.
↑
Regression, Classification, Detection,
Optical flow estimation, Motion prediction

Text-text Generation ✓
Text-image Generation ✓
img -img Generation ✓

• Unsupervised learning

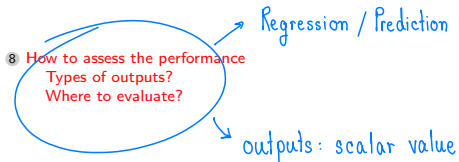
- Clustering algorithms
- PCA
- k-mean clustering
- Naive Bayes
- Gaussian mixture model
- Expectation-maximization algorithm (EM algorithm)

Dimensionality Reduction

↓
/ matching / Ranking / Reconstruction / Compression / Clustering

• DBSCAN / HDBSCAN.

↳ Density-Based Spatial Clustering of Applications with Noise



→ มีการ Evaluate ๒แบบ

What are the types of output to be measured?

- **Point estimate:** an output $\hat{y}$ from estimation which is a single value, i.e., $\hat{y} \sim \hat{Y}$:

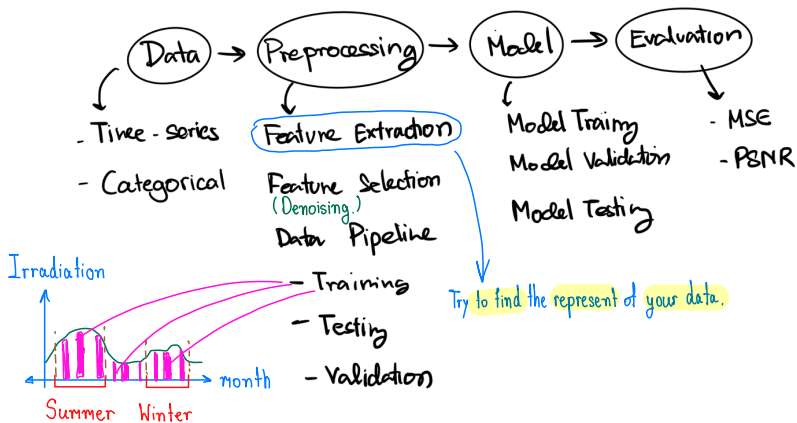
$$\hat{Y} = f(X) \quad \text{เราสนใจแบบนี้}$$

- **Interval estimate:** an output from estimation which is an interval with the probability (confidence level) of $1 - \alpha$, which is the probability that a true value y_o is in the interval $[\hat{y}^L, \hat{y}^H]$.

$$[\hat{y}^L, \hat{y}^H] \quad \text{such that} \quad P(y_o \in [\hat{y}^L, \hat{y}^H]) = 1 - \alpha$$

Where to measure?

- Quality of the predicted output y versus the ground truth y_o ^{no noise}
- Quality of the learned coefficients β versus the ground truth β_o



Homogenous dataset = การแบ่ง data set ให้มีลักษณะใกล้เคียงกัน.

- 9 Performance evaluation
 - Assessments for point estimator
 - Theoretical properties

Assessments for point estimator

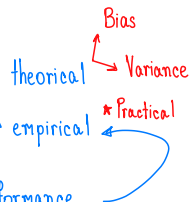
สำหรับ Discriminative Task ไม่ใช่ Regression.

- Measuring through the theoretical **properties** → MSE.
- Measuring through the **sampling** distribution

- Standard Errors (SE)
- Relative Standard Errors (RSE)
- Residual Sum of Square (RSS)
- R^2 Statistic (R^2)

Empirical Performance

Regression Tasks.



Theoretical properties

What is the performance of the point estimator $\hat{Y} = f(X)$ or $\hat{Y}_m = f(X_1, X_2, \dots, X_m)$, if the estimator received m input random samples? Machine Learning. Random Sample from pop X

Note that, we use y to denote **a reference value**. Prediction Value. a prediction m samples / m random sample.

We can measure the performance of the point estimator $\hat{Y}$ (or $\hat{Y}_m$) with the followings...

- ① **Bias** $\Rightarrow$ How far is the mean value of Y from a reference value y

$$\text{Bias}_y[\hat{Y}] = E[\hat{Y}] - \textcircled{y} \text{ no noise}$$

$$E[\hat{Y}]$$

$$\text{theoretical} = \int_{y \in \mathbb{C}} y \cdot f(y) dy$$

$$\text{empirical} = \sum_{i=1}^n y_i / n$$

$\leftarrow$ is a pdf as weight.
 $\leftarrow$ is a weight summing.

- ② **Efficiency** $\Rightarrow$ Suppose that an estimator $\hat{Y}$ is used for estimating y which is a model parameter associated with a probabilistic distribution of a random variable q , or $p(q; y)$. In fact, it is common that the estimator $\hat{Y}$ is also a function of $q_1, q_2, \dots, q_m$ observations that are sampled from $p(q; y)$ and are iid. The efficiency of the estimator $\hat{Y}$ can be defined as:

$$\text{Eff}_y(\Theta) = \frac{LB}{\text{Var}[\Theta]} = \frac{1}{m \mathcal{I}_{p(q; \theta)}(y) \text{Var}[\Theta]}$$

where $LB = \frac{1}{\mathcal{I}_{p(q_1, q_2, \dots, q_m; y)}(y)}$, and $\mathcal{I}_{p(q_1, q_2, \dots, q_m; y)}(y)$ is the Fisher information of the joint pdf $p(q_1, q_2, \dots, q_m; y)$. Then, $\mathcal{I}_{p(q_1, q_2, \dots, q_m; y)}(y) = m \mathcal{I}_{p(q; y)}(y)$ for iid observations. A good estimator should have the efficiency $\text{Eff}_y(\Theta)$ close to 1; the bad ones' are close to 0.

- ③ **Consistency** $\Rightarrow$ As $m \rightarrow \infty$, $\hat{Y}_m$ converges to a reference value y in probability.

$$\lim_{m \rightarrow \infty} P(|\hat{Y}_m - y| \leq 0)$$

****Note that these evaluations are based on a single reference value y .**

เกี่ยวกับ Notation

$$Y = y_0 + \varepsilon ; \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

↓
ค่าคงที่ noise จากการเก็บ data มา

ค่าที่บันทึกมา (measurement)

$$X = \text{input features} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ \vdots & \vdots & & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix}$$

x_j

$$X_i = [x_{i1}, x_{i2}, \dots, x_{in}]^T$$

i - # features = 1, 2, 3, ..., n

j = # data points = 1, 2, 3, ..., m

$$m \geq n$$

$E(X)$ → มองว่า X เป็นตัวแปรสุ่ม

Joint.

$E(X, Y)$ → "—" X, Y "—" พร้อมกัน.

Expectation (บางทีจะแปลเป็นแค่ E)

Remind

- ①. ในชีวิตจริงเราสามารถลด ε ได้
โดยขึ้นกับวิธี การเก็บข้อมูล และ transform
- ②. หอผ่าน ①. เราจะใช้ $Y = y_0 + \varepsilon$ ไปสร้าง ML
ซึ่งกระบวนการสร้าง ML เราจะจัดการ ε ได้แล้ว
และเราจะเรียกมันว่า Irreducible error.
- ③. โมเดลเราจะได้ $\hat{Y} = f(X)$ ซึ่งจะประมาณค่าได้
 Y ($\hat{Y} \approx y_0 + \varepsilon$) แต่ค่า error ของเราจะเรียก
 $\varepsilon = Y - \hat{Y}$ มาประมาณต่อ.

10 Bias-variance decomposition

Settings

Mean Square Error — 1. The stochastic behavior in our inputs X .

Mean Square Error — 2. The measurement noise $\mathcal{E}$

Bias-variance decomposition

Settings

Goal: the mean square error between the estimator and the measurements.

แต่ถ้าเราไม่รู้ Probabilistic Model

นี่แปลว่า เราสามารถใช้

Sample mean / Sample Variance

มาทำการคำนวณได้

Let

- X be the input information (an independent variable).
- Our estimator $\hat{Y} = f(X)$ predicts the label given the input information (this results in a dependent variable). **Thus, $\hat{Y}$ inherits the statistical properties from X .**
- **Reference:** y_o represents the ground truth of the estimation.
- But in **real settings**, instead of getting y_o , we may be able to access the measurements Y that is corrupted by noise: $Y = y_o + \mathcal{E}$. Usually, the measurement noise $\mathcal{E}$ is independent from X and has zero mean $\mathcal{E} \sim \mathcal{N}(0, \sigma^2)$.

X can behave randomly.

no noise (constant)

Random Variable

Biased-variance decomposition is derived from the mean square error:

$$E[f(X)] = E[\hat{Y}] \approx \frac{1}{n} \sum \hat{y}_j$$

Proof ในหน้า
32-35

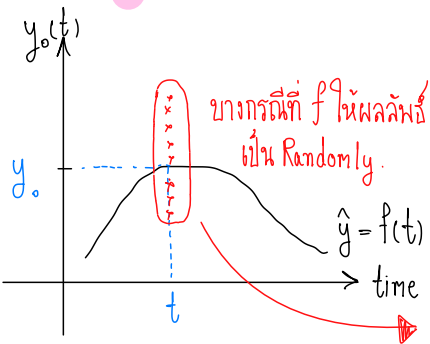
$$\begin{aligned} \text{MSE}_Y(\hat{Y}) &= E_{\hat{Y}, Y}[(\hat{Y} - Y)^2] \\ &= E_{X, Y}[(f(X) - Y)^2] \\ &= E_{X, \mathcal{E}}[(f(X) - y_o - \mathcal{E})^2] \\ &= \text{Bias}_{y_o}[f(X)]^2 + \text{Var}_X[f(X)] + \sigma^2 \\ &= (E_X[f(X)] - y_o)^2 + \{E_X[f(X)^2] - (E_X[f(X)])^2\} + (E[\mathcal{E}^2]) \end{aligned}$$

$Y = y_o + \mathcal{E}$ [Measurement = ground + noisy]

$E[\mathcal{E}^2] = \sigma^2$

$$\text{Bias}_{y_0}[f(x)] = E_x[f(x)] - y_0$$

reference.



กรณี 1 : f เป็น deterministic model

e.g. Linear Regression

เราใส่ค่า X ใดๆ ไปแล้วเราจะได้ $\hat{y}$ ออกมา 1 ค่า
ตลอด $\Rightarrow \text{Bias}_{y_0}[f(x)] = \hat{y} - y_0$ ไม่ต้องหา
expectation

กรณี 2 : f อื่นๆ (โมเดลมีความ Randomly)

เราจะได้ $f(x)$ หลายรูปแบบมากๆ เช่น
การ set seed , sample จากบาง distribution
ของ Probabilistic Model.

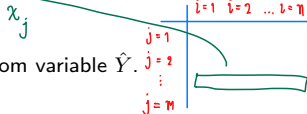
Deriving the bias-variance equation.

$$\text{MSE}(\hat{Y}) = \text{Bias}_{y_o}[f(X)]^2 + \text{Var}_X[f(X)] + \sigma^2$$

มองว่าเส้นค่าคงที่
↑ (ไม่ใช่ตัวแปรสุ่ม)
+ $\sigma_{\varepsilon}^2 = 0$

- 1: The **stochastic behavior in our inputs X** , where we assume that $\varepsilon = 0$.
For each i th sample $x_j \sim X$, the mapped value $\hat{y}_j$ will have different value depending on x_j :

$$\hat{y}_j = f(x_j)$$



Thus, the mapping from X will result in a dependent random variable $\hat{Y}$.

$$\hat{Y} = f(X)$$

This mean that if we have an estimator $f(X)$, it is possible that $f(X)$ may not completely aligned with Y , although there is no measurement noise. That is,

$$\hat{Y} \approx \boxed{Y = y_o} \quad \text{Ideal case.}$$

- 2: The **measurement noise ε** :

$$Y = y_o + \varepsilon.$$

Here, Y is the observed and noisy measurements. Then, the difference between our estimator and the reference y_o can be written as follows:

Real case $\rightarrow$

$$\hat{Y} = f(X) \approx Y = y_o + \varepsilon.$$

แปลว่าตอนเราสร้าง f
โมเดลค่ารวม Noise ไปด้วย.

- Note** $f(X)$ is a function of X and can be independent from the measurement noise ε . And it is often assumed that X is independent of ε .

Mean Square Error — 1. The stochastic behavior in our inputs X .

Setting: considering only the **stochastic behavior of X** .

เรากำลังมองว่า วิธีการเก็บข้อมูล
เราโคตรดี ดีจนไม่มี noise ($\mathcal{E} = 0$)

X is the random variable that captures the stochastic behavior of x . Our measurements $Y = y_0$ as $\mathcal{E} = 0$.

Mean square error (MSE) between $f(X)$ and Y is:

$$\begin{aligned} \text{MSE}(f) &= \mathbb{E}_{X,Y}[(f(X) - \underline{Y})^2] \\ &= \mathbb{E}_X[(f(X) - y_0)^2] \\ &= \underbrace{\text{Bias}_{y_0}[f(X)]^2}_{\text{Bias}} + \underbrace{\text{Var}_X[f(X)]}_{\text{Variance}} \end{aligned} \quad \begin{aligned} (1) \\ (2) \\ (3) \end{aligned}$$

constant neglect

$$\begin{aligned} \text{MSE}(\hat{\theta}) &= \mathbb{E}[(\hat{\theta} - \theta)^2] = \mathbb{E}\left[(\hat{\theta} - \mathbb{E}(\hat{\theta}) + \mathbb{E}(\hat{\theta}) - \theta)^2\right] \\ &= \mathbb{E}\left[(\hat{\theta} - \mathbb{E}(\hat{\theta}))^2 + 2(\hat{\theta} - \mathbb{E}(\hat{\theta}))(\mathbb{E}(\hat{\theta}) - \theta) + (\mathbb{E}(\hat{\theta}) - \theta)^2\right] \\ &= \mathbb{E}\left[(\hat{\theta} - \mathbb{E}(\hat{\theta}))^2\right] + 2\mathbb{E}\left[(\hat{\theta} - \mathbb{E}(\hat{\theta}))(\mathbb{E}(\hat{\theta}) - \theta)\right] + \mathbb{E}\left[(\mathbb{E}(\hat{\theta}) - \theta)^2\right] \\ &= \mathbb{E}\left[(\hat{\theta} - \mathbb{E}(\hat{\theta}))^2\right] + 2(\mathbb{E}(\hat{\theta}) - \theta)\mathbb{E}(\hat{\theta} - \mathbb{E}(\hat{\theta})) + (\mathbb{E}(\hat{\theta}) - \theta)^2 \\ &= \mathbb{E}\left[(\hat{\theta} - \mathbb{E}(\hat{\theta}))^2\right] + (\mathbb{E}(\hat{\theta}) - \theta)^2 \\ &= \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta}, \theta)^2 \end{aligned}$$

This is a constant, so the expectation of this term is just the term itself.
This term is a constant, so the expectation of this term is just the term itself.
This term is a constant, so the expectation of this term is just the term itself.

Notes:

- $\text{Bias}_{y_0}[f(X)] = \mathbb{E}_X[f(X)] - y_0$.
- $\text{Var}[f(X)] = \mathbb{E}[(f(X) - \mathbb{E}[f(X)])^2]$
- More complex $f(X) \rightarrow$ the **lower bias**, but **higher variance**. $\xrightarrow{\text{to be}}$ computational complexity.
- More complex $f(X) \rightarrow$ the higher flexibility in modeling, but also comes with a higher cost of computation.

Mean Square Error — 2. The measurement noise $\mathcal{E}$

Setting: we consider only the measurement noise $\mathcal{E} \rightarrow Y = y_o + \mathcal{E}$.

- Suppose that $X = x$ is just a constant.
- Thus, let $\epsilon = f(x) - y_o$ be the difference between $f(x)$ and y_o that is reducible depending on the choice of $f(x)$.

Mean square error (MSE) between $f(x)$ and Y is calculated as follows:
(also, do not forget that $y_o = Y - \mathcal{E}$)

$$\begin{aligned}\text{MSE}_Y(f) &= E_{\mathcal{E}}[(f(x) - Y)^2] \quad \text{constant.} \\ &= E_{\mathcal{E}}[(f(x) - (y_o + \mathcal{E}))^2] \quad \text{randomly.} \\ &= E_{\mathcal{E}}[(f(x) - y_o - \mathcal{E})^2] \\ &= E_{\mathcal{E}}[(\epsilon - \mathcal{E})^2] \\ &= E_{\mathcal{E}}[\epsilon^2 - 2\epsilon\mathcal{E} + \mathcal{E}^2]\end{aligned}$$

Suppose that the measurement noise $\mathcal{E}$ has zero mean and variance σ^2 . Then, the error becomes:

$$\begin{aligned}\text{MSE}(f) &= \epsilon^2 + \sigma^2 \quad \text{not reducible} \\ &= (f(x) - y_o)^2 + \sigma^2 \quad \text{high noisy data, sensor.}\end{aligned}$$

... σ^2 is irreducible error.

Bias-variance decomposition— putting Case 1 and Case 2 together

Setting: consider **both** the stochastic behavior of X and measurement noise $\mathcal{E} \rightarrow Y = y_o + \mathcal{E}$. Then ...

$$f(X) \approx Y = y_o + \mathcal{E} \sim \mathcal{N}(0, \sigma^2)$$

Mean square error (MSE) between $f(X)$ and Y becomes:

$$\begin{aligned} \text{MSE}(\hat{f}) &= \mathbb{E}_{X\mathcal{E}}[(f(X) - Y)^2] \\ &= \mathbb{E}_{X\mathcal{E}}[(f(X) - (y_o + \mathcal{E}))^2] \\ &= \mathbb{E}_{X\mathcal{E}}[((f(X) - y_o) - \mathcal{E})^2] \\ &= \mathbb{E}_{X\mathcal{E}}[(f(X) - y_o)^2 - 2\mathcal{E}(\hat{f}(X) - y_o) + \mathcal{E}^2] \\ &= \boxed{\mathbb{E}_X[(f(X) - y_o)^2]} - \cancel{2\mathbb{E}[\mathcal{E}]\mathbb{E}[f(X) - y_o]} + [\mathbb{E}_{\mathcal{E}}[\mathcal{E}^2] - (\mathbb{E}_{\mathcal{E}}[\mathcal{E}])^2] \\ &= \underbrace{\text{Bias}_{y_o}[f(X)]^2 + \text{Var}_X[f(X)]}_{\text{Reducible}} + \underbrace{\sigma^2}_{\text{Irreducible}} \end{aligned}$$

- $\mathbb{E}[\mathcal{E}] = 0$
- Above equation is known as **bias-variance decomposition**
- Again, σ^2 is irreducible error, no matter how well we construct $\hat{f}(x)$.
- Generally, the best model is the one achieving **the lowest bias and variance on testing data**. ★★★

MSE

Empirical $\rightarrow \bar{E}_n = \frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2$ $\xrightarrow{\hat{\mu}_\varepsilon}$ ทำไมเราถึงประมาณได้นะ?

Theory $\rightarrow \mathbb{E}[\varepsilon] = \mathbb{E}_{y, \hat{y}}[(\hat{y} - y)^2]$

$$\mu_\varepsilon = \text{Var}_X[f(X)] + (\text{Bias}_y[f(X)])^2 + \sigma^2$$

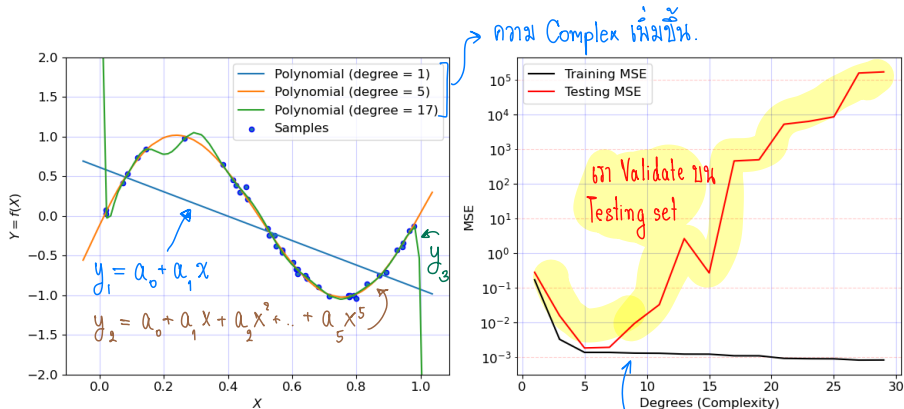
คำตอบ ∇ จาก Law of Large Number (LLN);

$$\lim_{n \rightarrow \infty} P(|\bar{E}_n - \mathbb{E}[\varepsilon]| > k) = 0 \text{ for } k > 0$$

เมื่อมีจำนวน data ที่มากพอ $\rightarrow$ ค่าทางปฏิบัติ จะใกล้เคียงกับ ค่าทางทฤษฎี

Original $\bar{X}_n = \frac{1}{n} \sum X_i$
 $P(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0$
as $n \rightarrow \infty$

Why Bias-Variance from MSE is important?



- Polynomial Regression with the high degree = 5, 17 can **flexibly** capture the details of the data better than degree = 1, so **less bias**.
- But **highly complex** $f(x)$ (with degree ≥ 6) $\Rightarrow$ **higher variance** (MSE).
- High flexibility** $\Rightarrow$ **less bias** but can also have **high variance**.
- Low flexibility** $\Rightarrow$ **high bias** but usually **low variance**.
- Note.** Poly degree 5 has the **lowest testing MSE** (also lower than degree 1).

เราไป plot MSE จะเห็นว่า
ไม่ว่าจะจริง MSE training จะลดลงเรื่อย ๆ (เพราะเรา learning ใช้นั้น fit มากขึ้น)
แต่ MSE testing จะไปต้นตึกเพิ่มตอนหลังๆ

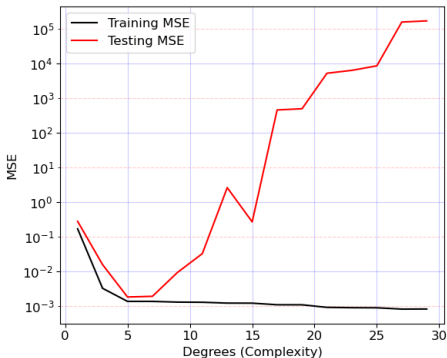
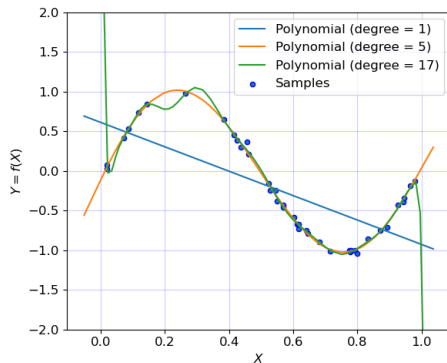
Why bias-variance from MSE is important?

From

$$\text{MSE}(\hat{f}) = \text{Bias}_{y_o}[f(X)]^2 + \text{Var}_X[f(X)] + \sigma^2$$

Reducible from ML Irreducible

It is possible that either bias or variance is dominating the overall MSE, which can be viewed jointly as a trade-off between the two values.



From above example (LHS), the model with higher complexity often accounts for the bias better, but if the complexity is too high, then the model can be overfitted to the training data. As such, the model is referred to as the model with *high flexibility*.

Why Bias-Variance from MSE is important?

- Model with high flexibility $\Rightarrow$ often high prediction accuracy.
- Model with low flexibility $\Rightarrow$ often high interpretability.

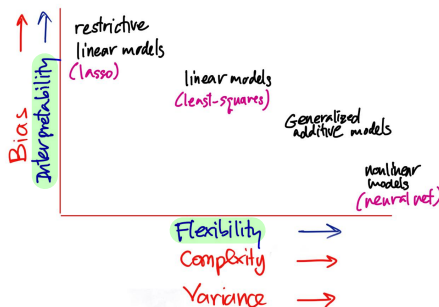


Figure: Trade-off between interpretability and accuracy [2, 4]

Measuring through sampling from distribution

→ ต้องมี Samples $X_i \Rightarrow$ แล้ววัดผล.

12 Measuring through sampling from distribution

Residual sum of square (RSS)

Standard error (SE)

Residual standard error (RSE)

R^2 Statistic

Residual sum of square (RSS)

Suppose that we have an estimator $\hat{Y}$ for the target measurement random variable Y .

- $\hat{y}_j, \dots, \hat{y}_m$ are sampled from the estimator $\hat{Y}$
- $y_j, \dots, y_m$ are sampled from the measurements Y ($Y = y_o + \mathcal{E}$).

Residual sum of square (RSS) is defined as the residuals between the samples of estimator and the target measurements:

$$\text{RSS} = \sum_{j=1}^m (\hat{y}_j - y_j)^2 \quad (4)$$

Handwritten annotations for the equation above:
- Red arrow from y_j to "measurement"
- Red arrow from $\hat{y}_j$ to "estimated output"
- Red arrow from the minus sign to "have noisy."

Recap ตัวแปร

$\hat{Y} = f(X)$	$\longleftrightarrow$	$\hat{y} = f(\underline{x})$
$Y = y_o + \mathcal{E}$	$\longleftrightarrow$	$y = y_o + \mathcal{E}$
Random		Constant.

Handwritten annotations for the recap table:
- Red arrow from X to $\underline{x}$ with label $x_1, x_2, x_3, \dots, x_1, x_2, \dots, x_i$
- Red arrow from $\mathcal{E}$ to $\mathcal{E}$ with label "ตัวแปรสุ่ม"
- Red arrow from $\mathcal{E}$ to $\mathcal{E}$ with label "ตัวแปรสุ่ม"

Standard error (SE)

Standard error (SE) of $\hat{Y}$ is the standard deviation of the samples $\hat{y} \sim \hat{Y}$.

$$SE(\hat{Y}) = E[(\hat{y} - \bar{y})^2] \rightarrow \underbrace{\frac{1}{m} \sum_{j=1}^m (\hat{y}_j - \bar{y})^2}_{\text{Empirical formula.}} \quad (5)$$

where

- $\hat{y}$ denote the samples of the estimator $\hat{Y}$.
- $\bar{y}$ is the expected value of $\hat{Y}$. $\bar{y} = E[\hat{Y}] \approx \frac{1}{m} \sum_j \hat{y}_j$

Relative standard error is defined as the ratio between the standard error (SE) and the expectation of the samples:

$$\text{Relative standard error}(\hat{Y}) = \frac{SE(\hat{Y})}{\bar{y}} * 100 \quad (6)$$

Sometimes, one might consider the relative standard error as it accounts for the magnitude of the sample.

Note that, by the definition, SE is a theoretical quantity. For example, the expectation operation requires some knowledge of the probability of the data samples $P(\hat{Y})$. In practice, we can calculate SE using a **bootstrapping technique** to compute $\bar{y} \approx \frac{\sum_j \hat{y}_j}{m}$.

Standard error (SE) in practice

Generally, if we can access $\hat{Y}$ directly, the calculation of SE is simple...

- Sample $\hat{y}_j$ for 1000 samples (as an example) from the entire population $\hat{Y}$
- Calculate the mean $\bar{y}$ of the sampled data
- Calculate the variance using the data samples and the the mean $\bar{y}$.


$$\bar{y} = \frac{1}{1000} \sum_{j=1}^{1000} \hat{y}_j$$

$$SE(\hat{Y}) = \frac{1}{1000} \sum (\hat{y}_j - \bar{y})^2$$

Standard error of sample mean estimation

For example, if we want to calculate SE of the sample mean for $\hat{Y}$, the steps are as follows...

- At first, we will use the entire population of $\hat{Y}$ to calculate the overall expectation $\bar{y}_{\text{overall}}$.
- Then, we will perform the bootstrapping trial for N repetition, and compute the sample mean for each trial.
 - Sample 1000 samples (as an example) from the entire population $\hat{Y}$, and calculate the sample mean $\bar{y}_j$ of the sampled data.
 - Repeat the calculation from $j = 1$ to m .
- Then, the standard error will be

$$SE = \frac{1}{m} \sum_{j=1}^m (\bar{y}_j - \bar{y}_{\text{overall}})^2$$

Residual standard error (RSE)

Residual standard error (RSE) measures the following quantity:

$$\text{RSE} = \sqrt{\frac{\text{RSS}}{m-p}} = \sqrt{\frac{\sum_{j=1}^m (y_j - \hat{y}_j)^2}{m-p}} \quad \begin{array}{l} m = \text{จำนวน data.} \\ p = \text{จำนวนพารามิเตอร์} \end{array} \quad (7)$$

where p is the number of parameters in the model. In regression model, the denominator is $m-p$ which is used instead of just m because $\sqrt{\frac{\text{RSS}}{m}}$ will under estimate the variance. For example, a simple linear regression is a linear function of X which is

$$\hat{Y} = \beta X + \beta_0 \quad \text{มี 2 พารามิเตอร์ } p=2.$$

Here the number of model parameter is 2. Note that, if we can get the prediction $\hat{Y} - Y = \epsilon$, then ϵ will be the only factor towards the error. The formulation to calculate RSE (where the denominator becomes $m-2$) is:

$$\text{RSE} = \sqrt{\frac{\text{RSS}}{m-2}} = \sqrt{\frac{\sum_{j=1}^m (y_j - \hat{y}_j)^2}{m-2}} \quad (8)$$

Normally, we use RSE to judge the value of the standard deviation of ϵ .

However, it is not always clear what constitutes a good RSE because RSE is measured in the unit of Y . Other statistics (R^2 and relative standard error) can be good alternatives.

R² Statistic

เกี่ยวข้องกับ Accuracy.

R² Statistic measures the lack of fit and the value is between 0 to 1.

measurement

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS} \quad (9)$$

$$RSS = \sum_{j=1}^m (y_j - \hat{y}_j)^2$$

$$TSS = \sum_{j=1}^m (y_j - \bar{y})^2$$

$TSS = \sum_{j=1}^m (y_j - \bar{y})^2$ measures the total variance of the samples of the measurements y_j from its mean ($\bar{y} = \sum_{j=1}^m \frac{y_j}{m}$). grand mean of measurement.

RSS defined as Eq.(4) measures the variability of the predicting model from the target measurements.

Then, $TSS - RSS$ measures the amount of variability in y_i after removing the variability from the predicting model. Therefore, the result from Eq.(9) will always have the value between 0 and 1.

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- [1] *Difference between Parametric and Non-Parametric Methods*.
<https://www.geeksforgeeks.org/difference-between-parametric-and-non-parametric-methods/>. Accessed: 2024-12-27.
- [2] Gareth James et al. *An Introduction to Statistical Learning: with Applications in R*. Springer, 2013. URL: <https://faculty.marshall.usc.edu/gareth-james/ISL/>.
- [3] Andrew Ng and Michael Jordan. "On Discriminative vs. Generative Classifiers: A comparison of logistic regression and naive Bayes". In: *Advances in Neural Information Processing Systems*. Ed. by T. Dietterich, S. Becker, and Z. Ghahramani. Vol. 14. MIT Press, 2001. URL: https://proceedings.neurips.cc/paper_files/paper/2001/file/7b7a53e239400a13bd6be6c91c4f6c4e-Paper.pdf.
- [4] Jitkomut Songsiri. "Statistical Learning". In: in EE575 Random Processes for EE. Chulalongkorn University, 2022. Chap. 1. Introduction to Statistical Learning. URL: <http://jitkomut.eng.chula.ac.th/ee575.html>.